FULL STACK WEB DEVELOPMENT LAB

Course Project

EduSync

Submitted by

Roll.No	Student Name	PRN
59	Aditya Yadav	1032232878
46	Apurva Herekar	1032232075
56	Manasvi Shelke	1032232227
47	Nirupam Surse	1032232545

Course Details

- **Programme: CSE Year:TY**
- **Course Name: Full Stack Web Development Lab**
- **Academic Year: 2025-26**
- **Semester: 6th**

Submitted To

Dr. Sanket Salvi
Faculty, Full Stack Web Development Lab

Date of Synopsis:
Date of Progress:
Date of Final Submission:

SECTION I: SYNOPSIS SUBMISSION

---

1. Abstract / Project Synopsis

EduSync Lite is a real-time collaborative web application designed to enhance peer-based academic learning. Students often rely on fragmented tools such as messaging apps and cloud drives to collaborate, which lack structured organization and real-time synchronization. This project proposes a unified platform where users can collaboratively create and edit study notes, upload resources, and interact in a shared digital workspace.

The system uses WebSockets for real-time synchronization, enabling instant updates across connected users. It incorporates user authentication, note management, file uploads, and basic role management. The project aims to demonstrate full-stack development concepts including REST APIs, real-time communication, database design, and secure authentication.

Expected outcomes include a fully functional, deployable collaborative platform that showcases system design, cloud-ready architecture, and modern frontend-backend integration techniques.

---

2. Introduction

2.1 Background and Motivation

With the increasing use of digital tools in education, collaborative learning has shifted online. However, students often use multiple platforms such as WhatsApp, Google Drive, and email, leading to disorganized workflows. There is a need for a unified collaborative platform specifically designed for academic group learning.

EduSync Lite is motivated by the need to create a structured, real-time learning environment that integrates collaboration, resource sharing, and communication within a single web application.

2.2 Problem Statement

Students lack a centralized platform that supports real-time collaborative note creation and structured resource management in an academic context. Existing tools are either too general-purpose or lack synchronization and organization features tailored to group study.

2.3 Objectives

- To design and develop a real-time collaborative web application.
- To implement secure authentication using JWT.
- To enable real-time note synchronization using WebSockets.
- To develop a structured resource upload and management system.
- To deploy a fully functional full-stack application.

---

3. Literature Review / Related Work

Existing systems such as Google Docs, Notion, and Trello provide collaboration features. However:

- Google Docs focuses primarily on document editing.
- Trello emphasizes task management.
- Notion provides structured knowledge management but is complex for simple academic use.

Most existing platforms are general-purpose and not tailored for lightweight student collaboration. EduSync Lite addresses this gap by offering a simplified, academic-focused collaboration system with real-time features and minimal complexity.

---

4. Proposed System

4.1 Proposed Solution

EduSync Lite provides a real-time collaborative board where authenticated users can:

- Create and edit notes.
- Upload and manage study resources.
- View live updates from other users.
- Track activity logs.

The system uses WebSocket communication for instant synchronization and REST APIs for structured data operations.

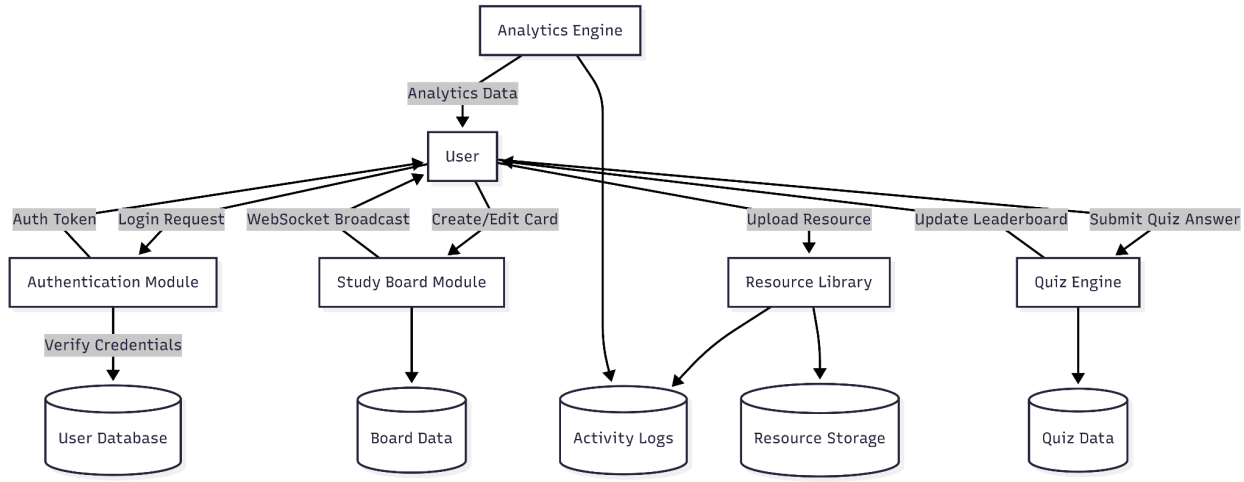
4.2 System Architecture (Proposed)

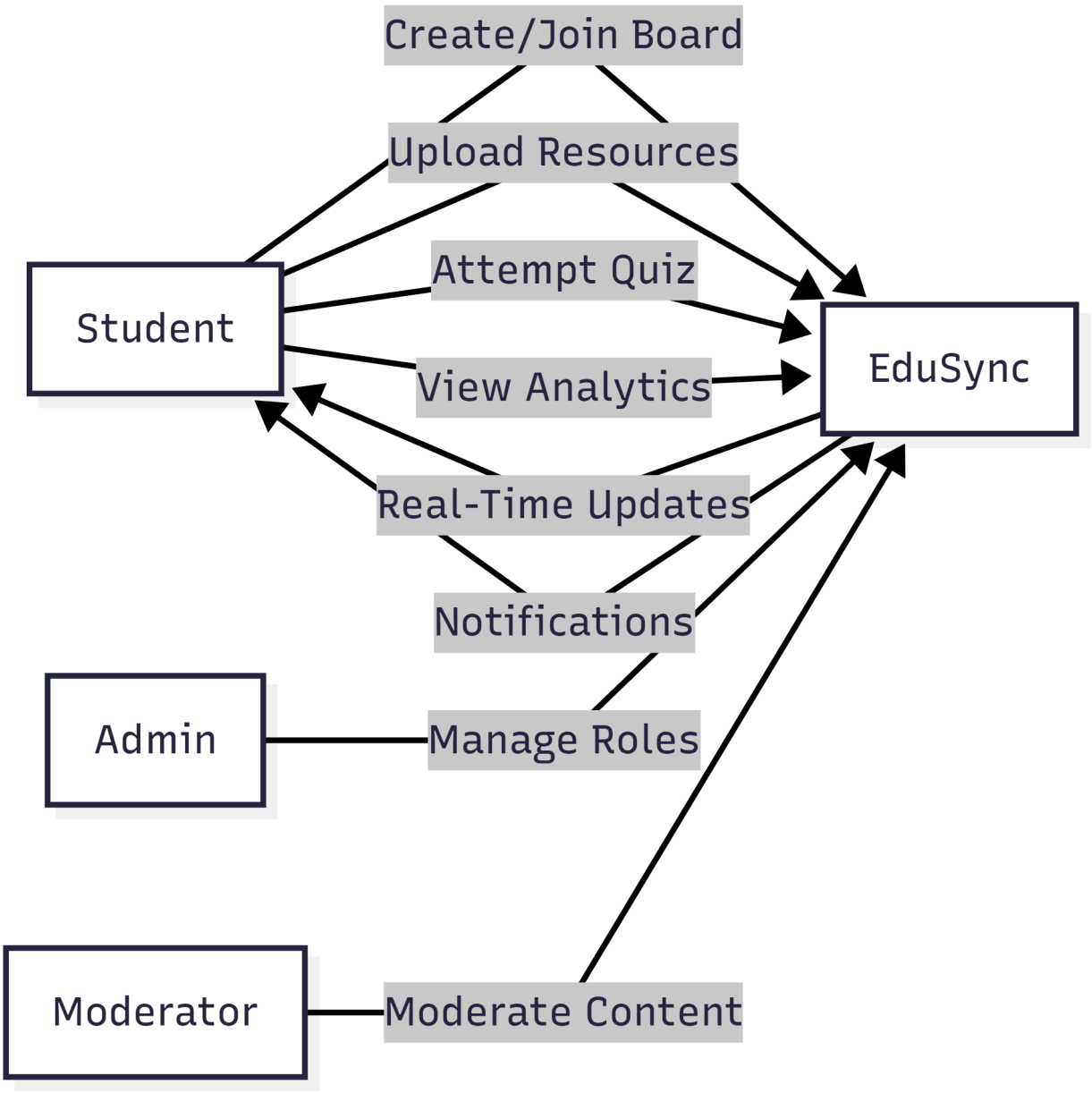
Architecture Components:

1. Frontend (React.js)
2. Backend (Node.js + Express + Socket.IO)
3. Database (PostgreSQL / SQLite)
4. File Storage (Local uploads / Cloud storage)

High-Level Flow:

User → React Frontend → REST API / WebSocket → Node Backend → Database → File Storage





4.3 Proposed Technology Stack

Frontend:

- React.js
- TailwindCSS
- Axios
- Socket.IO Client

Backend:

- Node.js
- Express.js
- Socket.IO
- JWT Authentication

Database:

- PostgreSQL / SQLite

Tools / Frameworks:

- Git & GitHub
- VS Code
- Postman
- Nodemailer

---

5. Scope of the Project

The project includes:

- User authentication and authorization.
- Real-time collaborative board.
- Resource upload and retrieval.
- Activity logging.
- Basic UI dashboard.

The project excludes:

- AI-based recommendations.
- Large-scale enterprise deployment.
- Mobile native application development.

---

6. Expected Outcomes

- Fully functional real-time collaborative web app.
- Secure login and authentication system.
- Real-time synchronization across multiple users.
- Organized resource management.
- Deployable and shareable web application.

---

7. Project Plan

Phase 1: Requirement Analysis & Design (Week 1)
Phase 2: Backend Development (Week 2)
Phase 3: Frontend Development (Week 3)
Phase 4: Integration, Testing & Deployment (Week 4)

SECTION II: PROGRESS EVALUATION

---

8. System Overview (Revised)

The system has been partially implemented with authentication and real-time note updates functioning correctly. Core backend APIs are operational, and database schema has been established.

---

9. Requirement Analysis

9.1 Functional Requirements

Implemented:

- User registration and login
- Create, edit, delete notes
- Real-time updates via Socket.IO
- Resource upload

Pending:

- Export functionality
- Advanced activity analytics

9.2 Non-Functional Requirements

- Performance: Real-time updates with minimal latency.
- Security: JWT-based authentication and data validation.
- Usability: Clean, responsive UI.
- Scalability: Modular backend architecture.

---

10. System Design

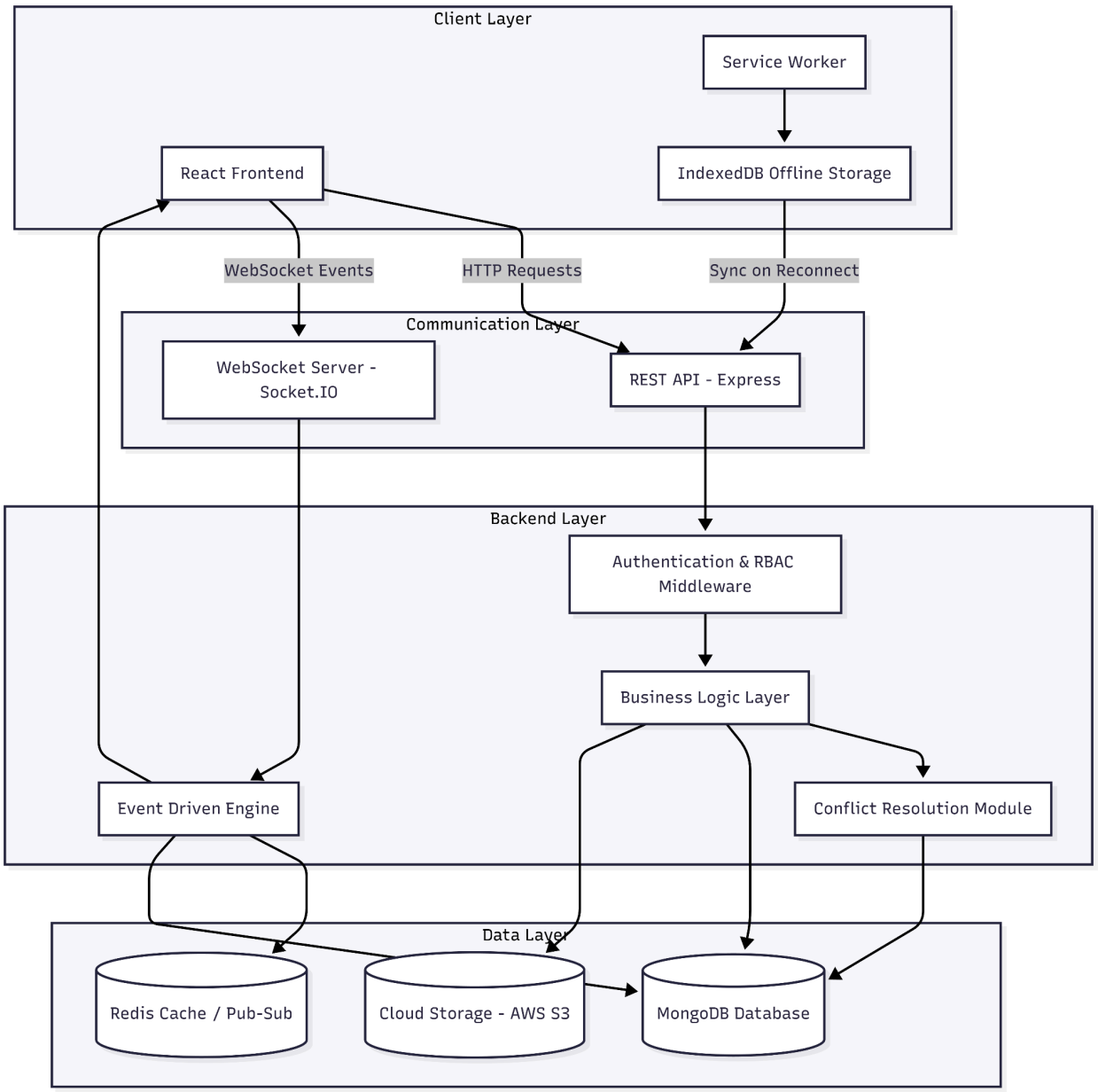
10.1 Use Case Diagram

Actors:

- User

Use Cases:

- Register/Login
- Create Note
- Edit Note
- Upload Resource
- View Board



10.2 Data Flow Diagram (DFD)

User → Frontend → API Server → Database
User → WebSocket → Backend → Broadcast to Connected Clients

10.3 Database Design

Tables:

Users (UserID, Name, Email, PasswordHash)
Notes (NoteID, Content, CreatedBy, Timestamp)
Resources (ResourceID, FilePath, UploadedBy, Timestamp)

---

11. Implementation Status

11.1 Frontend Development Status

- Login/Signup UI completed.
- Dashboard and board interface implemented.
- Resource upload UI integrated.

11.2 Backend Development Status

- Authentication APIs completed.
- CRUD APIs for notes implemented.
- WebSocket synchronization integrated.

11.3 Database Implementation Status

- Tables created.
- Relationships defined.
- Sample data inserted for testing.

---

12. Intermediate Results

- Real-time note synchronization functioning.
- File upload tested successfully.
- Multiple users can edit simultaneously.

---

13. Challenges Faced

- Handling real-time synchronization conflicts.
- Managing JWT authentication securely.

- Implementing file upload validation.

Mitigation:

- Used timestamps for conflict resolution.
- Middleware-based authentication.
- File type and size validation logic.

SECTION III: FINAL SUBMISSION

---

14. Final System Description

EduSync Lite is a fully functional real-time collaborative platform where users can log in, create and edit notes collaboratively, upload resources, and view live updates. The system ensures secure access and structured data management.

---

15. Implementation Details

15.1 Frontend Implementation

- Responsive UI using React and Tailwind.
- Form validation for authentication.
- Dynamic rendering of board notes.

15.2 Backend Implementation

- REST APIs for CRUD operations.
- JWT-based authentication.
- Socket.IO for real-time event broadcasting.

15.3 Database Implementation

- Relational schema implemented.
- Indexed primary keys.
- Data integrity constraints applied.

---

16. Testing

16.1 Test Plan

- Unit testing of APIs using Postman.
- Manual UI testing.
- Real-time multi-user testing.

16.2 Test Cases and Results

Example:

Test Case: User Login
Expected Result: Successful authentication
Actual Result: Passed

Test Case: Real-time Note Update
Expected Result: Update visible to all users
Actual Result: Passed

---

17. Results and Discussion

The system successfully demonstrates real-time collaboration. Performance remained stable under multiple user testing. The project effectively integrates frontend, backend, and database layers.

---

18. Deployment

Deployment Environment:

- Localhost / Render (Free Tier)

Backend hosted on:

- Node.js server

Frontend hosted on:

- Static hosting platform

---

19. Conclusion

EduSync Lite successfully achieves the objective of building a real-time collaborative web application. The project demonstrates strong understanding of full-stack development concepts including authentication, real-time communication, database design, and deployment.

---

20. Future Scope

- Add real-time chat module.
- Implement board export as PDF.
- Introduce role-based permissions.
- Add mobile PWA support.
- Scale using cloud infrastructure.

---

21. References

1. React Documentation – https://react.dev
2. Node.js Documentation – https://nodejs.org
3. Express.js Documentation – https://expressjs.com
4. PostgreSQL Documentation – https://www.postgresql.org/docs

5. Socket.IO Documentation – https://socket.io/docs

---

22. Appendix (Optional)

- **Source code snippets**
- **API endpoint list**
- **Sample screenshots**
- **User manual**